4.5.5 Angles in the same segment of a circle

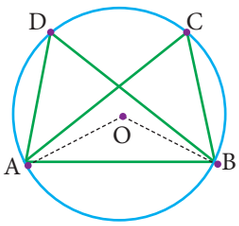


Fig. 4.74

Consider the circle with centre O and chord AB. C and D are the points on the circumference of the circle in the same segment. Join the radius OA and OB.

$$\frac{1}{2}\angle AOB = \angle ACB \quad \text{(by theorem 10)}$$

and $$\frac{1}{2}\angle AOB = \angle ADB \quad \text{(by theorem 10)}$$

$$\angle ACB = \angle ADB$$

This conclusion leads to the new result.

Theorem 11 Angles in the same segment of a circle are equal.

Example 4.9 In the given figure, O is the center of the circle. If the measure of $\angle OQR = 48°$, what is the measure of $\angle P$?

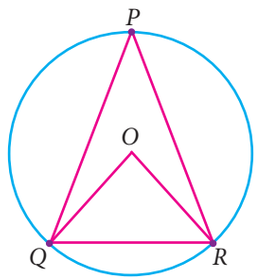


Fig. 4.75

Solution

Given $\angle OQR = 48°$.

Therefore, $\angle ORQ$ also is $48°$. (Why?_____)

$\angle QOR = 180° - (2 \times 48°) = 84°$.

The central angle made by chord QR is twice the inscribed angle at P.

Thus, measure of $\angle QPR = \frac{1}{2} \times 84° = 42°$.

Exercise 4.3

1. The diameter of the circle is 52cm and the length of one of its chord is 20cm. Find the distance of the chord from the centre.
2. The chord of length 30 cm is drawn at the distance of 8cm from the centre of the circle. Find the radius of the circle
3. Find the length of the chord AC where AB and CD are the two diameters perpendicular to each other of a circle with radius $4\sqrt{2}$ cm and also find $\angle OAC$ and $\angle OCA$.
4. A chord is 12cm away from the centre of the circle of radius 15cm. Find the length of the chord.

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